



Fig. 2: Examples of geometrical setting for the nervous system (left) and cardiac tissue (right) for $d = 2$.

We close the EMI problem with homogeneous Neumann boundary conditions:

$$(2.7) \quad \nabla u_i \cdot \mathbf{n}_i = 0 \quad \text{in} \quad \partial\Omega \cap \partial\Omega_i \quad \text{for} \quad i = 0, \dots, N.$$

Because of the pure Neumann boundary condition, the solving potentials u_i are defined up to an additive constant. Uniqueness can be enforced via Lagrange multipliers, pinning the solution in one point, or on the discrete level, providing a discrete nullspace to the solution strategy.

We investigate two cases of interest in the context of excitable tissues in physiological applications, cf. Figure 2, with final remarks on the general problem:

- (i) A relevant setting to model nervous system cells. In this case, a layer of extracellular media is always present between different cells (e.g. neurons and glial cells, see for example [43]), i.e. $\Gamma_i = \Gamma_{i,0}$ for all $i = 1, \dots, N$ or, equivalently, $\Gamma_{ij} = \emptyset$ if $i, j \neq 0$.
- (ii) A setting for cell-by-cell cardiac modeling. We consider the geometry of myocardiocytes, cardiac muscles cells which are connected by gap junctions or intercalated discs, with a somehow regular pattern [31].

3. Weak formulation and discrete operators. The EMI problem can be weakly formulated in various ways, depending on the unknowns of interest. We refer to [46] for a broad discussion on various formulations (including the so-called mixed ones). As it could be expected from the structure of (2.1)-(2.4), all formulations and corresponding discretizations give rise to block operators, with different blocks corresponding to Ω_i , and Γ_{ij} .

We use a so-called *single-dimensional* formulation and the corresponding discrete operators. In this setting, the weak form depends only on bulk quantities u_i since the current term I_{ij} is replaced by:

$$I_{ij} = \tau^{-1}(u_i - u_j + f_{ij}),$$

according to equations (2.3)-(2.4).

After substituting the expression for I_{ij} in (2.2), assuming the solution u_i to be sufficiently regular over Ω_i , we multiply the PDEs in (2.1) by test functions $\phi_i \in V_i(\Omega_i)$, with V_i a sufficiently regular Sobolev space with elements satisfying the boundary conditions in (2.7); in practice $V_i = H^1(\Omega_i)$ is a standard choice. After